

This is going to be a broad overview of the ideas and techniques used to prove the following much celebrated theorem.

Theorem (Gromov's Theorem on Groups of Polynomial Growth): If a finitely generated group Γ has polynomial growth, then Γ contains a nilpotent subgroup of finite index.

All of this work is discussed in Gromov's paper, and I am only writing this up for personal use (and to present to the University of Virginia Graduate Students Seminar).

Notions from Riemannian Geometry

Definition: Let $f: X \rightarrow Y$ be a map. We call such a map *globally expanding* if, for any two points $x_1 \neq x_2$ in X , one has $d(f(x_1), f(x_2)) > d(x_1, x_2)$.

We say f is *expanding* if each point $x \in X$ has a neighborhood U such that $f|_U$ is globally expanding.

Consider a compact connected Riemannian manifold, and let $f: X \rightarrow X$ be an expanding map. Then, X has no boundary. There is then a map $\tilde{f}: \tilde{X} \rightarrow \tilde{X}$ induced on the universal cover. Since any globally expanding map is a covering map on itself, it follows that the map $\tilde{f}$ is a homeomorphism, and the inverse map $\tilde{f}^{-1}: \tilde{X} \rightarrow \tilde{X}$ is a contraction.

Furthermore, for any $\delta > 0$, there is some $\varepsilon > 0$ such that for any $y_1, y_2 \in \tilde{X}$ with $d_{\tilde{X}}(y_1, y_2) \geq \delta$, one has

$$d_{\tilde{X}}(\tilde{f}^{-1}(y_1), \tilde{f}^{-1}(y_2)) \leq (1 - \varepsilon)d_{\tilde{X}}(y_1, y_2),$$

which follows by the definition of globally expanding. Therefore, $\tilde{f}^{-1}$ has a unique fixed point by the contraction mapping theorem, so projecting back down yields that $f: X \rightarrow X$ has a fixed point as well.

Two theorems of Shub and Franks are fundamental to the proof. To understand them, though, we require understanding a few results.

References

- [Gro81] Mikhael Gromov. "Groups of polynomial growth and expanding maps". In: *Inst. Hautes Études Sci. Publ. Math.* 53 (1981), pp. 53–73. ISSN: 0073-8301,1618-1913. URL: http://www.numdam.org/item?id=PMIHES_1981__53__53_0.
- [DW84] L. van den Dries and A. J. Wilkie. "Gromov's theorem on groups of polynomial growth and elementary logic". In: *J. Algebra* 89.2 (1984), pp. 349–374. ISSN: 0021-8693. DOI: [10.1016/0021-8693\(84\)90223-0](https://doi.org/10.1016/0021-8693(84)90223-0). URL: [https://doi.org/10.1016/0021-8693\(84\)90223-0](https://doi.org/10.1016/0021-8693(84)90223-0).